

One mole of a real gas is subjected to heating at constant volume from (P_1, V_1, T_1) state to (P_2, V_1, T_2) state. Then it is subjected to irreversible adiabatic compression against constant external pressure of P_3 atm, till the system reaches final state (P_3, V_2, T_3) . If the constant volume molar heat capacity of real gas is C_V , then the correct expression for ΔH from state 1 to state 3 is

- (A) $C_V (T_3 - T_1) + (P_3 V_1 - P_1 V_1)$ (B) $C_V (T_2 - T_1) + (P_3 V_2 - P_1 V_1)$
 (C) $C_V (T_2 - T_1) + (P_3 V_1 - P_1 V_1)$ (D) $C_P (T_2 - T_1) + (P_3 V_1 - P_1 V_1)$

Real gas,
 $U = f(T, V)$
 $dU = \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV$
 $= C_V dT$ (Zero)

$\Delta H_1 = \Delta U_1 + \Delta(PV)$
 $\Delta H_1 = C_V (T_2 - T_1) + V_1 (P_2 - P_1)$

Step-2 $Z=0$
 $\Delta U_2 = q + W_{irr}$
 $= -P_3 (V_2 - V_1)$

$\Delta H_2 = \Delta U_2 + \Delta(PV)$
 $= -P_3 (V_2 - V_1) + P_3 V_2 - P_2 V_1$
 $= +P_3 V_1 - P_2 V_1$

① (P_1, V_1, T_1) $\xrightarrow[\text{(real)}]{\text{const. vol.}}$ ② (P_2, V_1, T_2)
 ΔH_1
 $\Delta H = \Delta H_1 + \Delta H_2$
 $= C_V (T_2 - T_1) + V_1 P_2 - V_1 P_1$
 $= C_V (T_2 - T_1) + P_3 V_1 - P_1 V_1$

One mole of an ideal gas undergoes a reversible process in which the entropy of the gas changes with absolute temperature T as $S = aT + C_{v,m} \ln T$, where a is a positive constant. If $T = T_0$ at $V = V_0$ then the volume dependence of the gas on temperature in this process is

- (A) $T = T_0 + \ln V$ (B) $T = T_0 + \frac{R}{a} \ln \frac{V_0}{V}$ ☒ (C) $T = T_0 + \frac{R}{a} \ln \frac{V}{V_0}$ (D) $V = V_0 + \frac{a}{R} \ln \frac{T}{T_0}$

$$pV = nRT$$

$$n = 1 \text{ mol.}$$

$$\frac{p}{T} = \frac{R}{V}$$

For ideal gas, $S = aT + C_{v,m} \ln T$, when $V = V_0$, $T = T_0$, $V = f(T)$

$$dS = a dT + \frac{C_{v,m}}{T} dT \quad \text{--- (1)}$$

For an ideal gas,

$$\begin{aligned} dS &= \frac{\delta q_{rev}}{T} = \frac{dU - W_{rev}}{T} \\ &= \frac{dU}{T} - \frac{(-p dv)}{T} \\ &= \frac{dU}{T} + \frac{p}{T} dv. \end{aligned}$$

$$dS = \frac{C_v dT}{T} + \frac{R}{V} dv \quad \text{--- (2)}$$

$$\begin{aligned} \text{①} &= \text{②} \\ a dT + \frac{C_{v,m}}{T} dT &= \frac{C_{v,m} dT}{T} + \frac{R}{V} dv \\ \int_{V_0}^V \frac{dv}{V} &= \frac{a}{R} \int_{T_0}^T dT \\ \ln \frac{V}{V_0} &= \frac{a}{R} (T - T_0) \end{aligned}$$

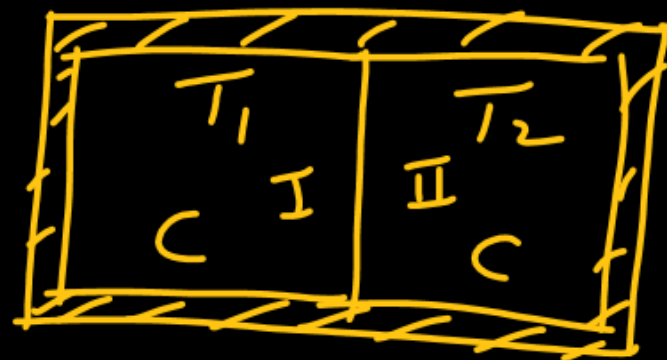
$$\boxed{\frac{R}{a} \ln \frac{V}{V_0} + T_0 = T}$$

Ans

Two blocks of copper metal are of the same size (heat capacity = C) but at different temperatures T_1 and T_2 . These blocks are brought together and allowed to attain thermal equilibrium. The entropy change of system is

- ✓ (A) $C \ln \left[\frac{(T_2 - T_1)^2}{4T_1 T_2} + 1 \right]$ (B) $C \ln \left[\frac{(T_2 - T_1)^2}{4T_1 T_2} \right]$ (C) $C \ln \left[\frac{(T_2 + T_1)^2}{4T_1 T_2} + 1 \right]$ (D) $C \ln \left[\frac{(T_2 + T_1)^2}{4T_1 T_2} - 1 \right]$

Solⁿ:



Thermal eq^m means finally temp. of both become equal ($= T_f$)
 → Assumption: No heat is lost or gained with surroundings.

Assume $T_1 > T_2$
 $\cancel{C(T_1 - T_f)} = \cancel{C(T_f - T_2)}$
 $T_f = \frac{T_1 + T_2}{2}$

$$\begin{aligned} \Delta S_T &= \Delta S_1 + \Delta S_2 \\ &= C \ln \frac{T_f}{T_1} + C \ln \frac{T_f}{T_2} \\ &= C \ln \frac{T_f^2}{T_1 T_2} = C \ln \left(\frac{(T_1 + T_2)^2}{4 T_1 T_2} \right) \\ &= C \ln \left(\frac{(T_1 - T_2)^2}{4 T_1 T_2} + 1 \right) \end{aligned}$$

The normal boiling point of a liquid is 350 K and ΔH_{vap} is 35 kJ/mol. Assume that ΔH_{vap} is independent from temperature and pressure. The correct statement(s) is/are

- Ans: A, B, D

Solⁿ: $(NBP)_{at\ 1\ atm} = 350\ K$, $\Delta H_{vap} = 35 \times 10^3\ J/mol$.

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$$(\Delta S)_{\text{vap, 1 atm}} = \frac{35 \times 10^3}{350} = 100 \text{ J/K/mol}$$



egⁿ. established

$P \downarrow$, eq. shifts forward

$$\Delta G = -ve \text{ (spontaneous)}$$

$$(\Delta G)_{\text{vap}} = 0 = \Delta H_{\text{vap}} - T_{\text{B.P.}} \times ((\Delta S)_{\text{vap}})_{\text{lat.}}$$

(C), ← (D)

$P = 2 \text{ atm}$

$P \uparrow$, e_L^m shifts backward.

$$\Delta G = \text{free} = \Delta H - T(\Delta S)$$

$(\Delta S)_{\text{un}}$

Two mol of a triatomic non-linear ideal gas is subjected to an adiabatic process from initial temperature 300 K and pressure 1 atm to a final pressure 16 atm. Find work done

(i) if process is reversible

(ii) If process is irreversible and external pressure of compression is 16 atm.

$$\begin{aligned} \text{(ii)} \quad W_{\text{irr}} &= \Delta U = nC_v(T_2 - T_1) \\ &= 2 \times 3R(425 - 300) \\ &= 6R \times 125 \\ &= 6750R \text{ J} \end{aligned}$$

Imp

Solⁿ: $n = 2 \text{ mol}$, $C_v = 3R$, $C_p = 4R$ } ignoring vibrational degrees of freedom.
 $T_1 = 300 \text{ K}$, $P_1 = 1 \text{ atm}$, $P_f = 16 \text{ atm}$.

(i) Rev. adi. process $\gamma = \frac{C_p}{C_v} = \frac{4}{3}$

Applicable only for rev. adi. process $q = 0$, $\Delta U = q + W_{\text{rev}}$
 $\Delta U = 0 + W_{\text{rev}}$
 $W_{\text{rev}} = \Delta U = nC_v(T_2 - T_1)$
 $= 2 \times 3R(T_2 - 300)$
 $\Rightarrow \frac{T_2}{T_1} = \left(\frac{P_1}{P_2}\right)^{\frac{1-\gamma}{\gamma}}$
 $PV^\gamma = \text{const}$
 $TV^{\gamma-1} = \text{const}$
 $T P^{\frac{1-\gamma}{\gamma}} = \text{const}$

$$\begin{aligned} &= \left(\frac{1}{16}\right)^{\frac{1-\frac{4}{3}}{\frac{4}{3}}} \\ &= \left(\frac{1}{16}\right)^{-\frac{1}{4}} \\ &= \frac{1}{(2^4)^{-\frac{1}{4}}} = 2 \\ \Rightarrow T_2 &= 2 \times 300 = 600 \text{ K} \\ W_{\text{rev}} &= 6R \times 300 = 1800R \text{ J} \end{aligned}$$

(ii) Irr. adi. process, $q = 0$
 $W_{\text{irr}} + 0 = \Delta U$
 $-P_{\text{ext}} \left(\frac{nRT_2}{P_2} - \frac{nRT_1}{P_1} \right) = nC_v(T_2 - T_1)$
 $-16 \left(\frac{T_2}{16} - \frac{300}{1} \right) = 3(T_2 - 300)$
 $4T_2 = 900 + 16 \times 300$
 $T_2 = 225 + 1200 = 1425 \text{ K}$

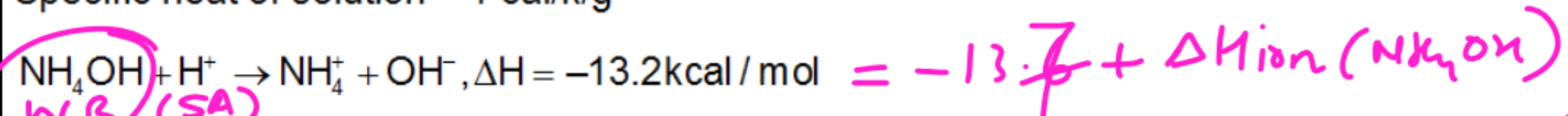
Imp

Estimate change in temperature when 100 ml of 0.1 M NaOH (aq) is added to 100 ml of 0.1 M NH_4Cl

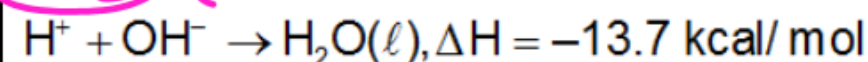
water equivalent of calorimeter = 50 g

Density of solution = 1 g/ml

Specific heat of solution = 1 cal/k/g



WB (SA)

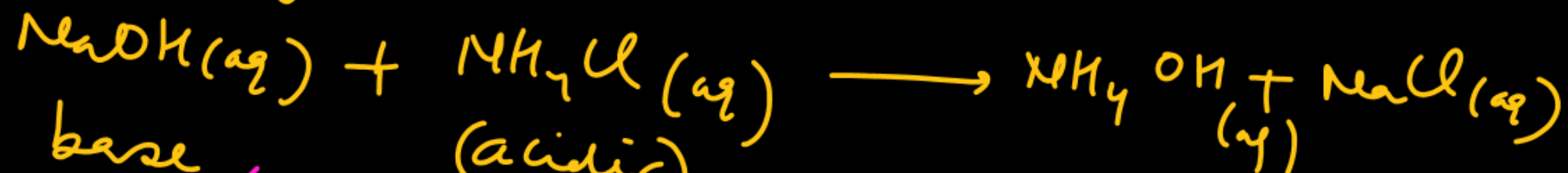


$\Rightarrow \Delta H_{\text{ion}}(\text{NH}_4\text{OH}) = 0.5 \text{ kcal/mol}$

→ In bomb calorimeter
↳ ΔU

→ In cont. pr. calorimeter
↳ ΔH

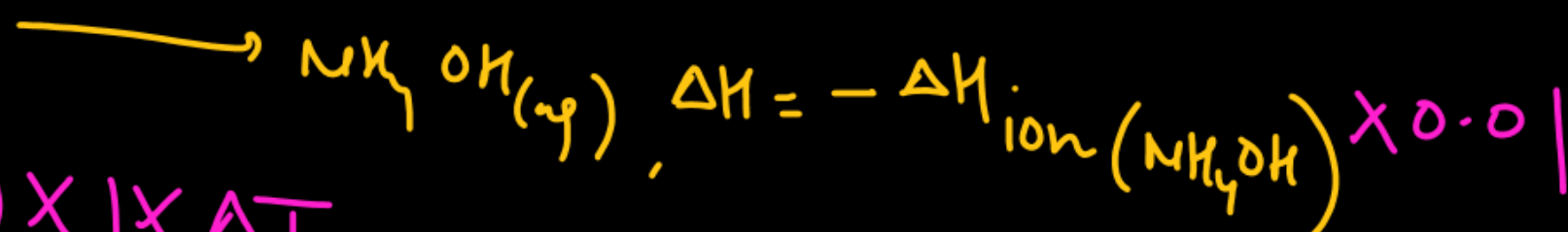
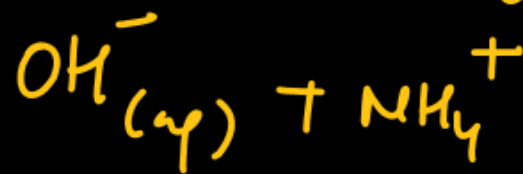
Solⁿ: Taking cont. pressure calorimeter.



base
0.01 mol

(acidic)

0.01 mol



$|\Delta H| = 5 = (200 + 50) \times 1 \times \Delta T$

$\Delta T = \frac{5}{250} = 0.02^\circ\text{C}$

$= -0.5 \times 1000 \text{ cal} \times 0.01$

$= -5 \text{ cal}$ (Heat liberated)

Temp. rise.

Three moles of an ideal gas ($C_{P,m} = 2.5R$) and 2 moles of another ideal gas ($C_{P,m} = 3.5R$) are taken in a vessel and compressed reversibly and adiabatically. In this process, the temperature of gaseous mixture increased from 300 K to 400 K. The increase in internal energy of gaseous mixture (in cal) is

Solⁿ: 3 mol, $C_{p,m} = 2.5R$ + 2 mol, $C_{p,m} = 3.5R$

Rev. adiabatic compression.
300 K $\longrightarrow$ 400 K.

$$(\Delta U)_{\text{mix}} = ? \quad \left| \quad \begin{array}{l} q_{\text{rev}} = 0 \\ \Delta U = q + w \\ \Delta U = w_{\text{rev}} \end{array} \right.$$

$$(\Delta U)_{\text{mix}} = n(C)_{\text{mix}} (T_2 - T_1)$$

$$= 5 \times 9.5R \times (400 - 300)$$

$$= 950R$$

$$= 250 \times 2 = 1900R \quad \text{Ans}$$

$$(C)_{\text{mix}} = \frac{n_1 C_1 + n_2 C_2}{(n_1 + n_2)}$$

$$= \frac{3 \times 2.5R + 2 \times 3.5R}{(3 + 2)}$$

$$= \frac{7.5R + 7R}{5} = \frac{14.5}{5}R$$

$$C_p - C_v = R$$

$$C_{v,m} = \frac{14.5}{5}R - R = \frac{9.5R}{5}$$

Assume ideal gas behaviour for all the gases considered and neglect vibrational degrees of freedom. Separate equimolar samples of Ne, O₂, CO₂ and SO₂ were subjected to two process as mentioned. Initially all are at same state of temperature and pressure.

Step I → All undergo reversible adiabatic expansion to attain same final volume, which is double the original volume thereby causing the decreases in their temperature.

Step II → After step I and are given appropriate amount of heat isochorically to restore the original temperature.

Make the correct option(s):

Ans: A, C, D

$v \rightarrow \text{constant}$

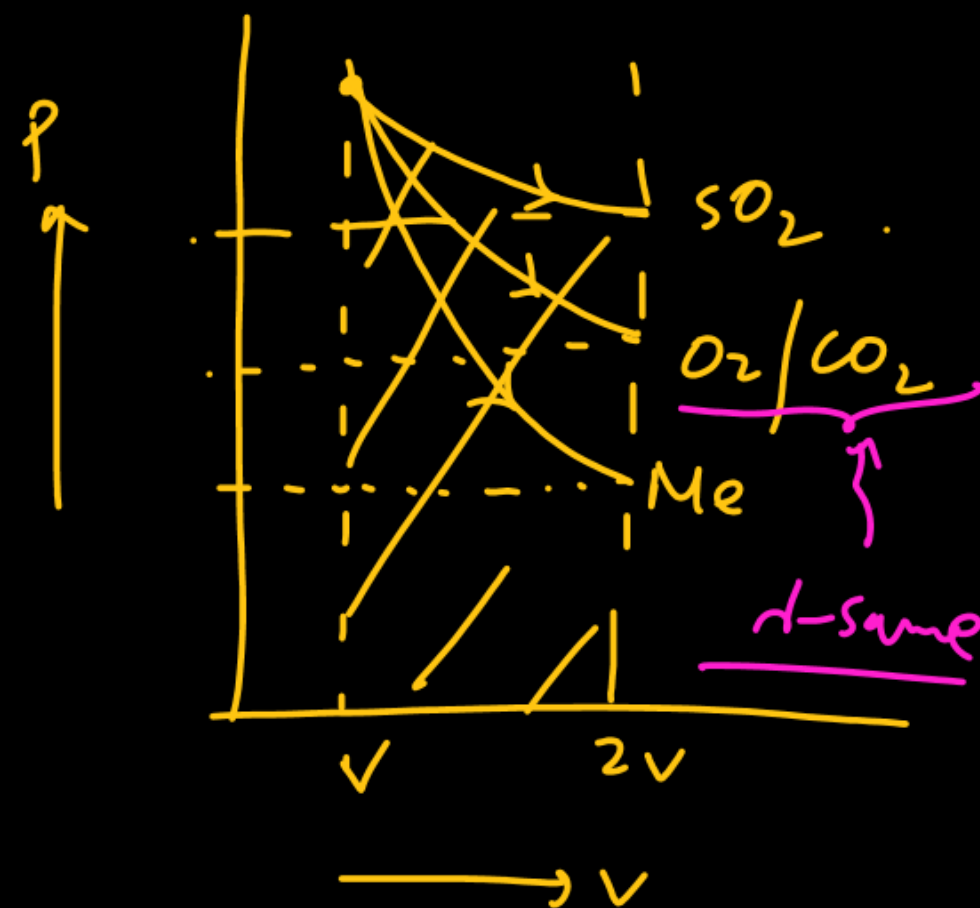
✓ (A) Due to step I only, the decrease in temperature will be maximum for Ne

✗ (B) During step II, heat given will be minimum for SO₂

✓ (C) There will be no change in internal energy for any of the gas after both the steps of process are completed

✓ (D) The P-V graph of O₂ and CO₂ will be same

V-Imp



Solⁿ:
 $\left. \begin{array}{l} \text{Ne, } \gamma = 1.67 \\ \text{O}_2, \gamma = 1.4 \\ \text{CO}_2, \gamma = 1.4 \\ \text{SO}_2, \gamma = 1.33 \end{array} \right\}$

Step-I, $PV^\gamma = \text{const}$, Temp. drop is max. for Ne.
 $\Delta U_1 = 0 + W_{\text{rev}}$ max. for SO₂.
 $|\Delta U_1| \rightarrow \text{max. for SO}_2$.
 I-step
Step-II, $\Delta U_2 = Q_2 + W_2$, $W_2 = 0$, $\Delta U_2 = Q_2$.
 $Q_2 = \Delta U_2$ (max. for SO₂).
 $\Delta U = \Delta U_1 + \Delta U_2 = 0$ (initial & final temp same).
 $\Delta U = 0$ for ideal gas.

When 1.0 g of oxalic acid ($\text{H}_2\text{C}_2\text{O}_4$) is burned in a bomb calorimeter whose heat capacity is 8.75 kJ/K , the temperature increases by 0.312 K . The enthalpy of combustion of oxalic acid at 27°C is :

- (A) -245.7 kJ/mol (B) -244.452 kJ/mol (C) -246.947 kJ/mol (D) none of these

Solⁿ: In Bomb Calorimeter, $q = \Delta U$, $(V \rightarrow \text{constant})$ / $\text{H}_2\text{C}_2\text{O}_4$ & H_2O are liquid at 27°C .

